

Supplementary Material for A Study of the Design Space of Motion and Disocclusion Control in Video Generation

In this supplementary material, we provide additional details on the user study, the implementation of DaS-Interp., and the prompt generation procedure. The uploaded supplementary folder (Supplementary) contains an annotation tool demonstration video and a webpage (Supplementary/Results/results_web.html) presenting all benchmark samples, including the ground-truth videos, tracking videos, disocclusion masks, prompts, and videos generated by each evaluated method. We also provide the Google Form used for the user study, together with anonymized response statistics, in the folder Supplementary/User_study.

Appendix A: User Study.

To complement the quantitative evaluation, we conduct a perceptual user study using pairwise preference comparisons. In each comparison, a participant is shown two videos generated by different methods for the same scene and asked to select the preferred one.

For the *synthetic scenes*, we randomly select 10 scenes from the full evaluation set. We compare all five methods against each other, resulting in $\binom{5}{2} = 10$ unique method pairs. Each scene is randomly assigned to one pair, so every method appears in exactly 4 comparisons. For the *captured scenes*, all 5 scenes are used, with each scene assigned to one randomly selected method pair such that every method appears in exactly 2 comparisons. Across both settings, each participant evaluates all 15 comparisons, with the two videos in each pair presented in random order to avoid position bias.

The participant were asked, for each pair, to indicate the best result according to (i) *Motion Match* (how closely the video follows the object motion shown in the ground truth video), (ii) *Disocclusion Match* (how well the video matches the target specification in the revealed areas), (iii) *Realism* (overall visual quality, including natural-looking shadows, minimal seams/artifacts, and smooth, consistent motion over time). The study takes between 5 and 10 minutes to complete. For each method, we report the winning rate –the proportion of votes won out of all votes cast in comparisons involving that method.

Answers. Figure A1 details the distribution of winning rate collected over 20 participants.

The user study results are largely consistent with our quantitative evaluation. For **motion quality**, tracking-based methods are generally preferred over text-only approaches. In the synthetic scenes, DaS-Interp. achieves the highest motion preference rate, while DaS performs best in the captured scenes, confirming the benefit of explicit motion guidance.

For **disocclusion quality**, image-conditioned methods are consistently preferred. DaS-Interp. receives the highest score in the synthetic scenes, while CogX-Interp. achieves an overwhelming majority of votes in the captured scenes. This aligns with our quantitative results and highlights the importance of explicit appearance references for preserving the identity and fidelity of newly revealed content.

The **realism** results are more complex. In the synthetic scenes,

DaS-Interp. receives the highest realism preference, suggesting that accurate motion and disocclusion contribute positively to overall visual quality. In contrast, CogX-Interp. and DaS are preferred in the captured scenes, indicating that perceived realism depends not only on motion and disocclusion accuracy, but also on appearance consistency and artifact levels. Overall, realism appears to be determined by a balance between motion quality, disocclusion fidelity, and visual coherence.

Appendix B: DaS-Interp. Implementation Detail

Since DaS-Interp. builds directly on DaS, we first briefly review the underlying DaS framework before describing our modification.

DiffusionAsShader

DaS [GYL*25] is an image-to-video latent diffusion model. It takes as input an image, a text prompt, and a 3D tracking video, and denoises a random noise into a realistic video expressing the desired motion. Since it is a latent diffusion model, both the image and the tracking video ($\mathbf{V}_{tr} \in \mathbb{R}^{(1+T) \times H \times W \times 3}$) are first converted into latent features using a frozen VAE encoder $\mathcal{E}$: $\mathcal{F}_{tr} = \mathcal{E}(\mathbf{V}_{tr}) \in \mathbb{R}^{(1+\frac{T}{4}) \times \frac{H}{8} \times \frac{W}{8} \times 16}$ (see their Fig. 2). Specifically, the input image is first preprocessed with zero padding to obtain the pseudo video $\mathbf{V}_s = \mathbb{R}^{(1+T) \times H \times W \times 3}$, which is further converted into a latent feature:

$$\mathcal{F}_s = \mathcal{E}(\mathbf{V}_s) \in \mathbb{R}^{(1+\frac{T}{4}) \times \frac{H}{8} \times \frac{W}{8} \times 16}. \quad (1)$$

The tracking video encodes the precise motion of the object of interest, while the input image provides the appearance reference for the diffusion model.

In the denoising process, a denoising Diffusion Transformer (DiT) ϵ_θ iteratively denoises an initial random noise z_N into a clean latent z_0 , conditioned on the text prompt and the encoded image and tracking video. The denoised latent is finally decoded into a resulting video $\mathbf{V} = \mathcal{D}(z_0)$. The denoising process at step n ($0 < n \leq N$) is defined as:

$$\epsilon_n = \epsilon_\theta(z_n, n, \mathbf{T}, \mathcal{F}_s, \mathcal{F}_{tr}), \quad z_{n-1} = z_n - \epsilon_n. \quad (2)$$

DaS-Interp.

In DaS-Interp., we fuse the input image and the target last frame in the latent space during DaS’s denoising process to produce the final motion- and disocclusion-aware video. Specifically, we concatenate the input image $\mathbf{I}_0$ and the edited target last frame $\mathbf{I}_{edit}$ along the temporal dimension and encode them using the frozen VAE encoder $\mathcal{E}$, yielding latent representations $\mathbf{z}_0$ and $\mathbf{z}_{edit}$. To match the temporal length of the video latent, we place $\mathbf{z}_0$ at the first temporal position and $\mathbf{z}_{edit}$ at the final temporal position, while filling all intermediate positions with zero latents:

$$\mathbf{Z}_c = [\mathbf{z}_0, \mathbf{0}, \dots, \mathbf{0}, \mathbf{z}_{edit}] \in \mathbb{R}^{13 \times 120 \times 180 \times 16},$$

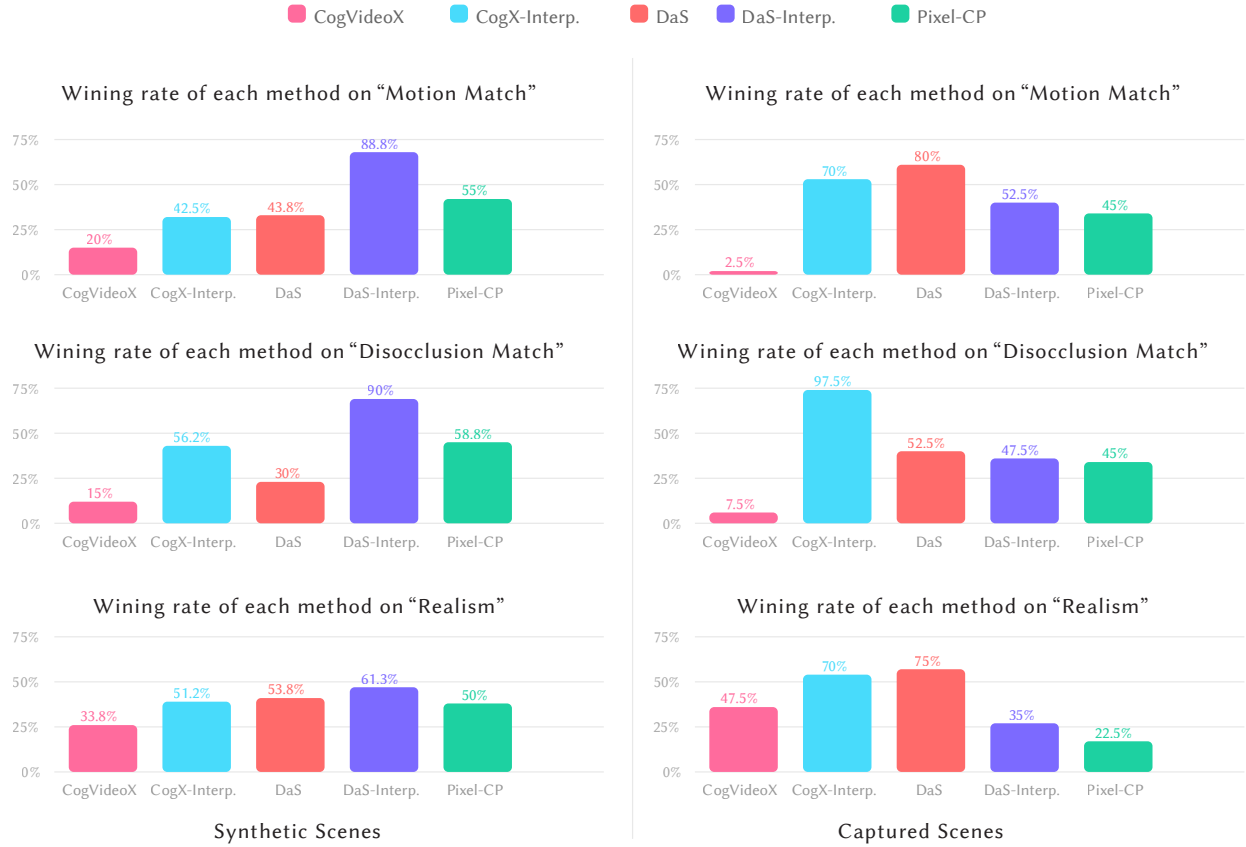


Figure A1: User study results. Win rates for motion quality, disocclusion quality, and realism across the five evaluated methods in the synthetic and captured scenes.

This conditioning tensor has the same temporal resolution as the video latent and replaces the original image-conditioning tensor used by CogVideoX. During denoising, the noisy video latent is concatenated with $\mathbf{Z}_c$ along the channel dimension and passed to the transformer together with the text embeddings. In parallel, we retain the tracking-video guidance mechanism of DaS to constrain the motion of foreground objects. The first-frame latent anchors the generated video to the input image, while the last-frame latent provides explicit image-based guidance for the appearance of disoccluded content. After iterative denoising, the resulting latent video is decoded by the VAE decoder $\mathcal{D}$ to produce the final video $\mathbf{V}$.

Appendix C: Prompt Generation Procedure.

The following prompt is provided to ChatGPT 5.5 [Ope26] to generate the motion, appearance, and simplified prompts for each ground truth video in the benchmark:

“I am creating a benchmark for motion and disocclusion control in image-to-video generation.

Given the uploaded video, generate the following three prompts:

- **Motion Prompt:** A detailed description of the motion of the visible foreground objects, including motion type (e.g., translation, rotation, deformation), direction, and relative movement. Describe

only the motion of objects visible in the first frame. Do not mention the disoccluded object or newly revealed content.

- **Appearance Prompt:** Describe the object or content that becomes visible due to the foreground object’s motion. Explain where it is initially hidden, how it is gradually revealed, and any motion it exhibits after becoming visible.
- **Simplified Prompt:** A concise description combining the primary motion event and the disocclusion event in a single sentence.